Name:
Registration:
A

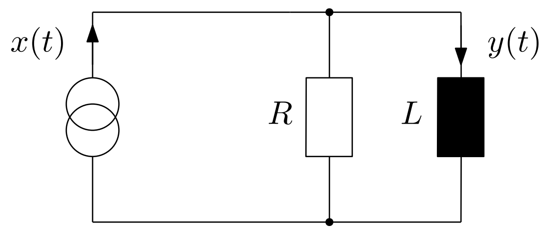
EE320 Signals & Systems Quiz

November 6, 2024

Duration: 9:05pm – 9.50am (45 minutes).
This midterm exam counts 5% towards your final mark of EE320.
Please read the instructions carefully before attempting ALL questions.
A formula sheet is provided, and you may use a dictionary. No other material is permitted.
Good luck!

---

1. (a) Derive a differential equation to describe the input-output behaviour of the following circuit, where a current source provides an with input current $x(t)$ and we want to know the output current $y(t)$:



A: Node analysis:

$$x(t) = \frac{1}{R}u(t) + y(t)$$

where $y(t)$ **is the current through the inductor, and** $u(t)$ **the voltage drop over either resistor or capacitor. Mesh analysis**

$$u(t) = L\dot{y}(t),$$

enables a substitution once the node equation:

$$x(t) = \frac{R}{L}\dot{y}(t) + y(t) \; .$$

(b) The elements now have values $C = 1$ and $R = 1$. Confirm that the transfer function for the circuit in 1(a) is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+1} \; ,$$

where $Y(s) \bullet\!\!-\!\!\circ\; y(t)$ and $X(s) \bullet\!\!-\!\!\circ\; x(t)$ are the Laplace transforms of the input and output of the circuit in Q1a).

A:

$$x(t) = \dot{y}(t) + y(t)$$

Laplace transform:

$$X(s) = (1+s)Y(s)$$

or

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+1} \; .$$

(c) Derive the time domain signal $y(t) \circ\!\!-\!\!\bullet\, Y(s)$ for the excitation with a step function, $x(t) = s(t)$, and sketch $y(t)$ on the interval $t \in (-1, 5)$.

A: Starting from Laplace via partial fraction expansion:

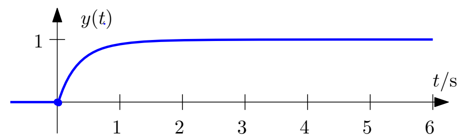
$$Y(s) = \frac{1}{s+1}\frac{1}{s} = \frac{-1}{s+1} + \frac{1}{s}$$

Hence

$$y(t) = (1 - \mathrm{e}^{-t})s(t)\ .$$

Some values:

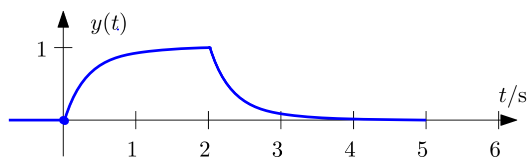
t	<0	**0**	**1**	∞
$y(t)$	**0**	**0**	$1 - \frac{1}{\mathrm{e}} \approx \frac{2}{3}$	**1**



(d) If the input is changed to $x(t) = s(t) - s(t-2)$, what is the new output? Please provide a rough sketch.

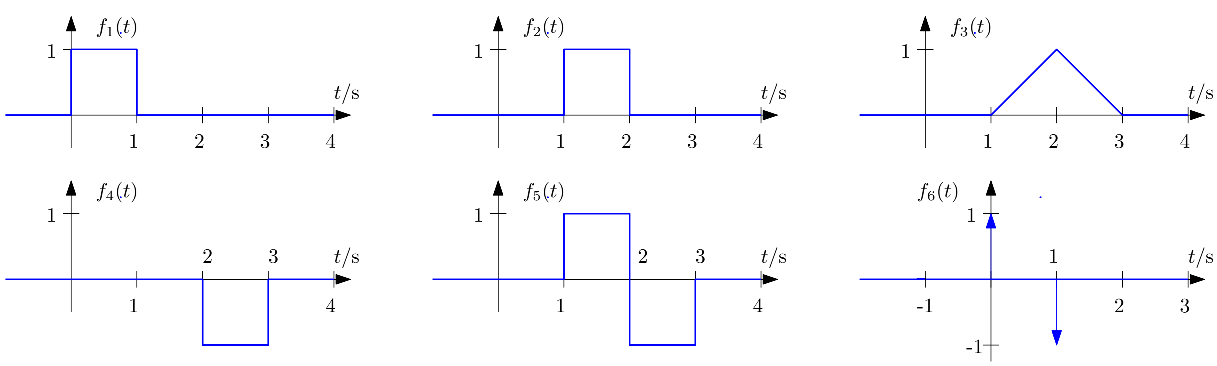
A: Since this is an LTI system, we apply superposition. The output must be

$$y_2(t) = y(t) - y(t-2) = (1 - \mathrm{e}^{-t})s(t) - (1 - \mathrm{e}^{-t+2})s(t-2)\ .$$



3 marks

2. (a) For the functions shown below, verify that $f_1(t) * f_2(t) = f_3(t)$.



A: Ideally this would be done graphically for a couple of values of t:

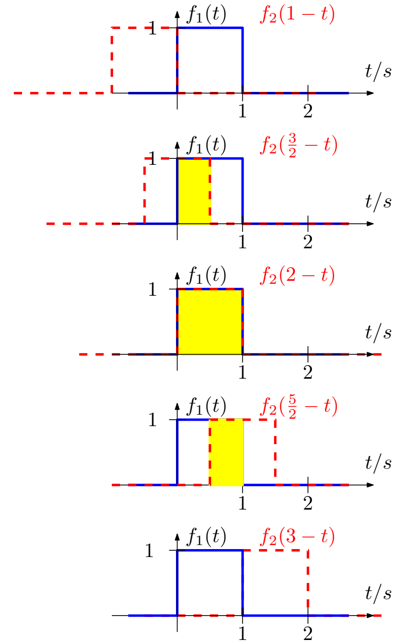
$$y_1(1) = \int_{-\infty}^{\infty} f_1(\tau) f_2(1-\tau) d\tau = 0$$

$$y_1(\frac{3}{2}) = \int_{-\infty}^{\infty} f_1(\tau) f_2(\frac{3}{2}-\tau) d\tau = \int_{0}^{\frac{1}{2}} d\tau = \frac{1}{2}$$

$$y_1(2) = \int_{-\infty}^{\infty} f_1(\tau) f_2(3-\tau) d\tau = \int_{0}^{1} d\tau = 1$$

$$y_1(\frac{5}{2}) = \int_{-\infty}^{\infty} f_1(\tau) f_2(\frac{5}{2}-\tau) d\tau = \int_{\frac{1}{2}}^{1} d\tau = \frac{1}{2}$$

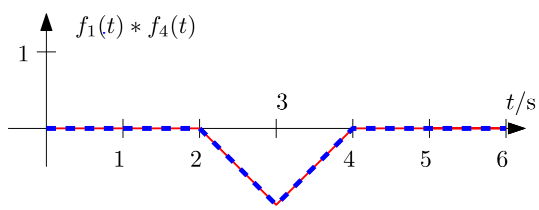
$$y_1(3) = \int_{-\infty}^{\infty} f_1(\tau) f_2(3-\tau) d\tau = 0$$



(b) Determine or justify the results of the convolution $f_1(t) * f_4(t)$.

A: Recognising that $f_4(t) = -f_2(t-1)$, we have

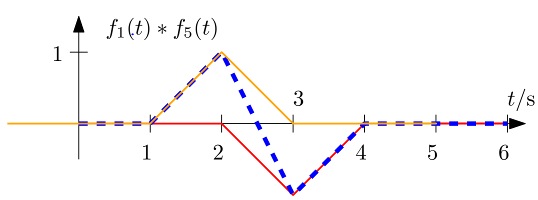
$$f_1(t) * f_4(t) = (-1) f_1(t) * f_2(t-1) = -f_3(t-1) \; .$$



(c) Determine or justify the result of $f_1(t) * f_5(t)$.

A: Recognising that $f_5(t) = f_2(t) + f_4(t)$, we apply superposition:

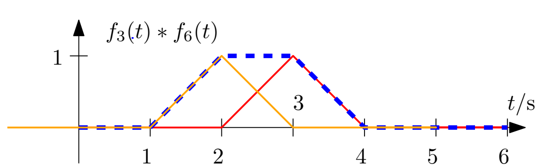
$$f_1(t) * f_5(t) = f_1(t) * \{f_2(t) + f_4(t)\} = f_3(t) - f_3(t-1) \; .$$



(d) Determine or justify the result of $f_3(t) * f_6(t)$.

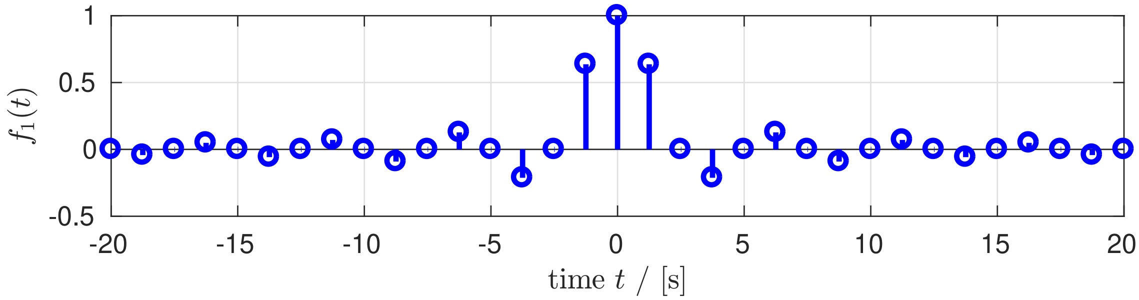
A: Again, best to apply superposition:

$$f_3(t) * f_6(t) = f_3(t) * \delta(t) + f_3(t) * \delta(t-1) = f_3(t+1) + f_3(t-1) \; .$$



3 marks

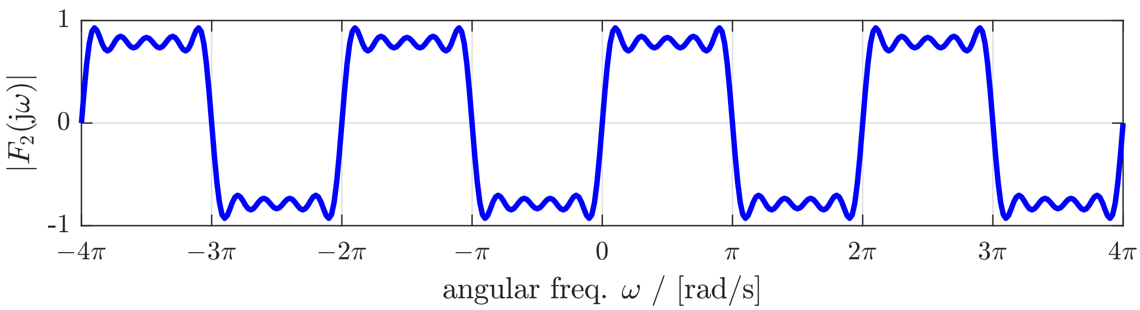
3. (a) What can you say about the Fourier transform $F_1(j\omega)$ •—○ $f_1(t)$ regarding periodicity and discreteness?



A: Time and Fourier domain equivalence $f_1(t)$ ○—• $F_1(j\omega)$:

$f_1(t)$	$F_1(j\omega)$
aperiodic	**continuous**
discrete ($T_s = 1$s)	**periodic ($f_s = 1$Hz)**

(b) What can you say about the signal $f_2(t)$ ○—• $F_2(j\omega)$ regarding periodicity and discreteness?



A: Fourier and time domain equivalence $F_2(j\omega)$ •—○ $f_2(t)$:

$F_2(j\omega)$	$f_2(t)$
periodic ($w_s = 2\pi$ of $f_s = 1$Hz)	**discrete ($T_s = 1$s)**
continuous	**aperiodic**

2 marks

End of Quiz.

Name:
Registration:
B

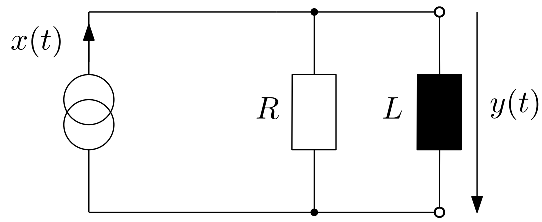
EE320 Signals & Systems Quiz

November 6, 2024

Duration: 9:05pm – 9.50am (45 minutes).
This midterm exam counts 5% towards your final mark of EE320.
Please read the instructions carefully before attempting ALL questions.
A formula sheet is provided, and you may use a dictionary. No other material is permitted.
Good luck!

---

1. (a) Derive a differential equation to describe the input-output behaviour of the following circuit, where a current source provides an with input current $x(t)$ and we want to know the output voltage $y(t)$:



A: Nodal analysis:

$$x(t) = \frac{1}{R}y(t) + i(t)\ .$$

To remove the auxiliary current $i(t)$ that flows through L, we use a mesh equation (or the voltage-current law of the inductor):

$$y(t) = L\frac{\mathrm{d}i(t)}{\mathrm{d}t} \qquad \rightarrow \qquad \frac{\mathrm{d}i(t)}{\mathrm{d}t} = \frac{1}{L}y(t)\ .$$

After differentiating the nodal equation:

$$\dot{x}(t) = \frac{1}{R}\dot{y}(t) + \frac{1}{L}y(t).$$

(b) The elements now have values $L = 1$ and $R = 1$. Confirm that the transfer function for the circuit in 1(a) is (potentially up to a minus sign)

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s}{s+1}\ ,$$

where $Y(s) \bullet\!\!-\!\!\circ\ y(t)$ and $X(s) \bullet\!\!-\!\!\circ\ x(t)$ are the Laplace transforms of the input and output of the circuit in Q1a).

A: PDE:

$$\dot{x}(t) = y(t) + \dot{y}(t)$$

Laplace transform:

$$sX(s) = (1+s)Y(s)$$

or

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s}{s+1}\ .$$

(c) Derive the time domain signal $y(t) \circ\!\!-\!\!\bullet\, Y(s)$ for the excitation with a step function $x(t) = s(t)$, and sketch $y(t)$ on the interval $t \in (-1, 5)$.

A: Laplace: $x(t) = s(t) \circ\!\!-\!\!\bullet\, \frac{1}{s}$**; therefore**

$$Y(s) = \frac{s}{(s+1)}\frac{1}{s} = \frac{1}{s+1}$$

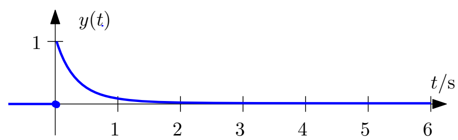
Hence

$$y(t) = \mathrm{e}^{-t}s(t) \ .$$

Some values:

t	< 0	**0**	**1**	∞
$y(t)$	**0**	**1**	$\frac{1}{\mathrm{e}^1} \approx \frac{1}{3}$	**0**

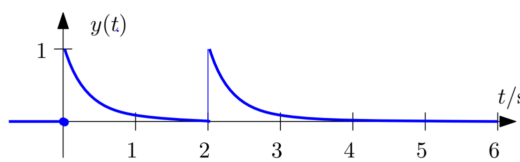
Sketch:



(d) If the input is changed to $x(t) = s(t) + s(t-2)$, what is the new output? Please provide a rough sketch.

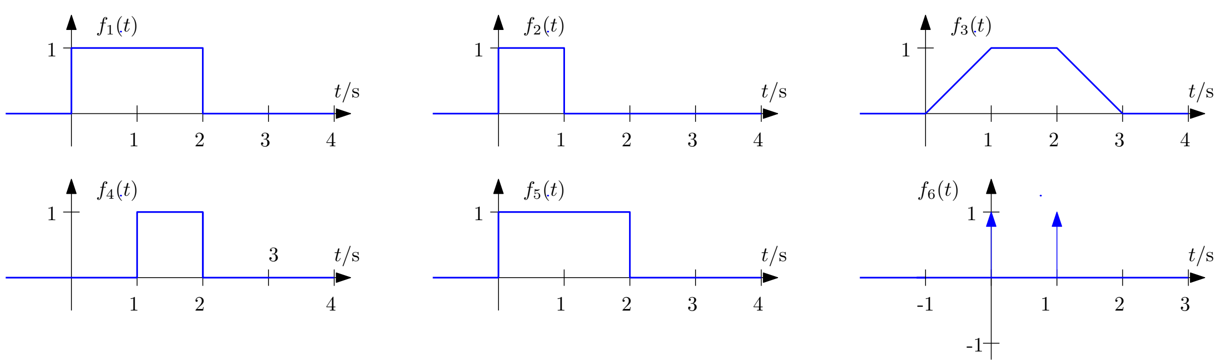
A: Since this is an LTI system, the output must be

$$y_2(t) = y(t) + y(t-2) = \mathrm{e}^{-t}s(t) + \mathrm{e}^{-t+2})s(t-2) \ .$$



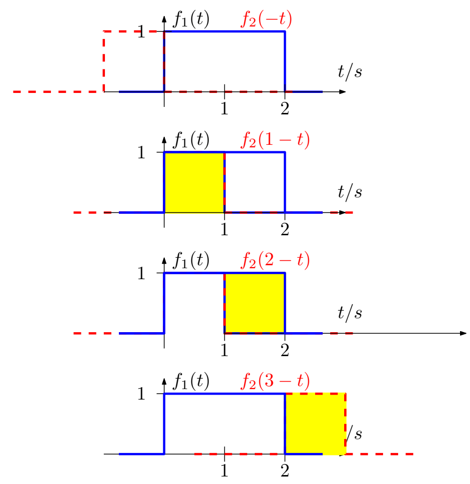
3 marks

2. (a) For the functions shown below, verify that $f_1(t) * f_2(t) = f_3(t)$.



A: Ideally this would be done graphically for a couple of values of t — generally convolving two pulses should yield a trapezoidal results:

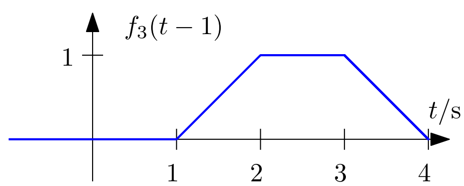
$$
\begin{aligned}
y_1(0) &= \int_{-\infty}^{\infty} f_1(\tau) f_2(0-\tau) d\tau = 0 \\
y_1(1) &= \int_{-\infty}^{\infty} f_1(\tau) f_2(1-\tau) d\tau = \int_{0}^{\frac{1}{2}} d\tau = 1 \\
y_1(2) &= \int_{-\infty}^{\infty} f_1(\tau) f_2(2-\tau) d\tau = \int_{0}^{1} d\tau = 1 \\
y_1(3) &= \int_{-\infty}^{\infty} f_1(\tau) f_2(3-\tau) d\tau = \int_{\frac{1}{2}}^{1} d\tau = 0
\end{aligned}
$$



(b) Determine or justify the results of the convolution $f_1(t) * f_4(t)$.

A: Note that $f_4(t) = f_2(t-1)$. Therefore

$$f_1(t) * f_4(t) = f_1(t) * f_2(t-1) = f_3(t-1) \ .$$

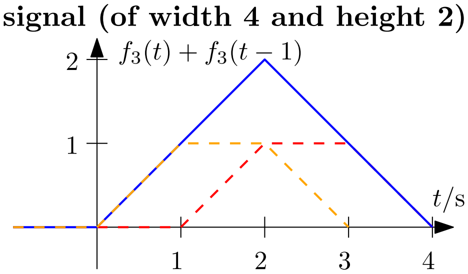


(c) Determine or justify the result of $f_1(t) * f_5(t)$.

A: Best to apply superposition: $f_5(t) = f_2(t) + f_4(t)$

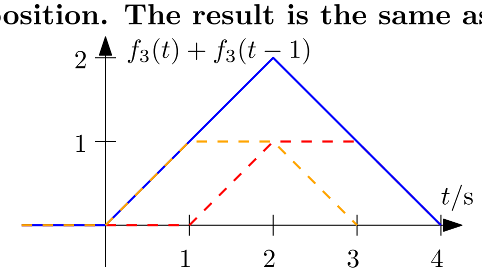
$$f_1(t) * f_5(t) = f_1(t) * \{f_2(t) + f_2(t-1)\} = f_3(t) + f_3(t-1) \ .$$

This is now a triangular signal (of width 4 and height 2).



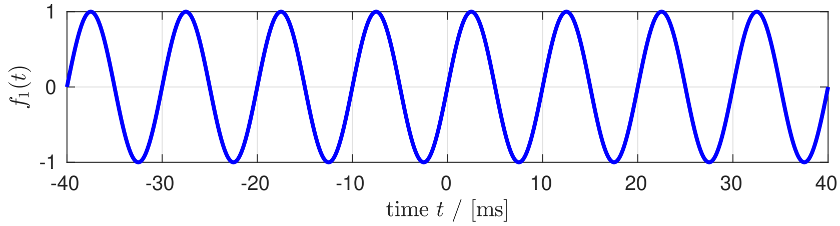
(d) Determine or justify the result of $f_3(t) * f_6(t)$.

A: Best to apply superposition. The result is the same as for (c).



3 marks

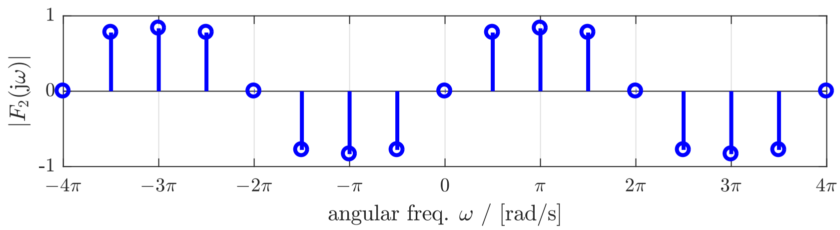
3. (a) What can you say about the Fourier transform $F_1(j\omega)$ •—○ $f_1(t)$ regarding periodicity and discreteness?



A: Time and Fourier domain equivalence $f_1(t)$ ○—• $F_1(j\omega)$:

$f_1(t)$	$F_1(j\omega)$
periodic, $T_0 = 10$**ms**	**discrete,** $f_0 = 100$**Hz**
continuous	**aperiodic**

(b) What can you say about the signal $f_2(t)$ ○—• $F_2(j\omega)$ regarding periodicity and discreteness?



A: Fourier and time domain equivalence $F_2(j\omega)$ •—○ $f_2(t)$:

$F_2(j\omega)$	$f_2(t)$
periodic ($\omega_s = 2\pi$, $f_s = 1$**Hz**)	**discrete** ($T_s = 1$s)
discrete ($\omega_0 = \frac{\pi}{2}$, $f_0 = \frac{1}{4}$**Hz**	**periodic** ($T_0 = 4$s)

2 marks

End of Quiz.